

Cross-Subject Hand Gesture Recognition via Inter-modal Alignment of sEMG and A-mode Ultrasound for Subject-Invariant Robotic Control

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Abstract—Surface electromyography (sEMG) and A-mode ultrasound (AUS) provide complementary electrophysiological and morphological information for hand gesture recognition. However, cross-subject deployment remains challenging because inter-subject variability and inter-modal heterogeneity cause substantial distribution shifts between training and target users. This paper proposes a Subject-Invariant Alignment and Fusion Network (SIAF-Net) for label-free cross-subject sEMG-AUS gesture recognition. SIAF-Net integrates bidirectional cross-modal attention for sEMG-AUS feature interaction with domain-adversarial learning for subject-invariant representation learning. The framework is evaluated on eight subjects using a leave-one-subject-out protocol, achieving a 6-class classification accuracy of $81.42\% \pm 7.65\%$. In online closed-loop robotic validation, SIAF-Net requires no labeled target-user gesture calibration and achieves an online accuracy of 82.14%, a task success rate of 87.50%, and a command jitter rate of 1.52%. The measured end-to-end response delay is 1.04 s, mainly limited by mechanical actuation and the sensing window, while the core algorithmic inference requires only 12.4 ms. These results demonstrate the feasibility of sEMG-AUS fusion for robust cross-subject robotic gesture interfaces.

I. INTRODUCTION

Gesture recognition plays a crucial role in decoding human movement intentions, serving as the core foundation for physical human-robot interaction, dexterous prosthetic control, and robotic teleoperation [1]. Since human hand movements involve complex neuromuscular dynamics, enabling machines to accurately and stably interpret these intentions has long been a significant challenge. Among various physiological signals, surface electromyography (sEMG) has become the most widely used modality in gesture recognition due to its non-invasiveness and high temporal resolution [2], [3]. In feature engineering, deep learning methods such as long short-term memory (LSTM) networks and convolutional neural networks (CNNs) have been extensively employed to directly extract global and local spatiotemporal features from sEMG signals [4], [5], [6]. However, sEMG is inherently a fragile electrophysiological signal, highly susceptible to interference from external physical factors such as skin impedance changes, sweat, and electrode displacement, re-

sulting in a low signal-to-noise ratio (SNR). Consequently, relying solely on the single modality of sEMG struggles to meet the stringent control stability requirements of modern robotic systems. To break through the perceptual bottleneck of single modalities, multi-modal fusion technology has been introduced to provide complementary semantic information. Many recent studies have attempted to combine sEMG with A-mode ultrasound (AUS) signals, which can capture changes in the internal physical morphology of muscles, to improve recognition robustness. For example, Zou et al. [7] fused sEMG and muscle force signals through specific spatial mapping techniques; Geng et al. [8] and Zeng et al. [9] also conducted in-depth explorations at the feature fusion level. These studies have demonstrated that the fusion of sEMG and AUS can achieve high offline recognition accuracy within specific subjects (intra-subject), far exceeding the performance of single modalities. However, when models are applied to target users without labeled gesture calibration, their performance often suffers substantial degradation. As shown in Fig. 1, when different subjects perform the exact same “scissors” gesture, their signal features exhibit significant individual differences; the data distribution differences between subjects are even greater than those between gesture classes. At the same time, the modal heterogeneity between sEMG and AUS further exacerbates the complexity of the feature space. How to effectively integrate heterogeneous multi-modal data while overcoming inter-individual distribution differences has become a key pain point in the current field of physiological signal-based gesture recognition.

The core bottleneck of cross-subject physiological signal recognition lies in the inconsistency of data distributions across individuals, namely the classic “domain shift” problem in machine learning. To alleviate this issue, unsupervised domain adaptation (UDA) techniques have been widely applied in the analysis of electroencephalography (EEG) and sEMG [10], [11]. Among them, domain adversarial neural networks (DANN) and their core component—the gradient reversal layer (GRL)—are currently the mainstream paradigm for achieving physiological signal distribution alignment [12]. In gesture recognition, Ketyk’o et al. [13] demonstrated that deep transfer learning and adversarial training can significantly improve the generalization performance of sEMG across different subjects; Chan et al. [14] also used adaptive networks to mitigate performance degradation caused by electrode displacement and subject differences. However, existing cross-subject alignment methods are mostly designed for single modalities [15] or only perform shallow multi-modal concatenation [16], failing to

This work was supported in part by the National Natural Science Foundation under Grant 62573401, in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 62403453, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grants 2025A1515010392 and 2025A1515011973.

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simultaneously and effectively suppress both “modal structural heterogeneity” and “individual distribution drift” during sEMG-AUS fusion. To address these challenges, this paper proposes an integrated “modality-subject” alignment framework, namely **Subject-Invariant Alignment and Fusion Network (SIAF-Net)**. SIAF-Net extracts subject-invariant representations by learning shared cross-modal semantics between sEMG and AUS. First, raw signals are transformed into physically interpretable features to mitigate initial spatial misalignment. A Bi-directional Cross-Attention mechanism then enables electrophysiological and morphological features to mutually calibrate, extracting robust complementary information. Crucially, we integrate a Domain Adversarial Neural Network (DANN) with a Gradient Reversal Layer (GRL) to reduce subject-specific “fingerprints.” This unified framework achieves mutual enhancement: cross-modal attention stabilizes feature drift for adversarial training, while domain adaptation regularizes the fusion process to uncover transferable biomechanical patterns across individuals. Although methods like Transformer Fusion [17] improve fusion, they fail to address anatomical shifts in AUS signals, leading to performance drops in cross-subject scenarios. Our framework tackles this by jointly optimizing modal and subject alignment. Our main contributions are:

1. To our knowledge, this work provides one of the first systematic evaluations of AUS and sEMG fusion for cross-subject gesture recognition. We have constructed and open-sourced a multimodal cross-subject sEMG-AUS gesture dataset containing 8 subjects.

2. An end-to-end alignment-and-fusion framework, SIAF-Net, is proposed. This framework captures the semantic correlations between electrophysiology and physical morphology through a bi-directional cross-modal attention mechanism, and couples this module with a domain adversarial mechanism.

3. A hardware-in-the-loop end-to-end perception and control pipeline is implemented. Experimental results show that, without labeled target-user gesture calibration, SIAF-Net supports online closed-loop robotic validation for held-out target users.

II. DATA COLLECTION AND PREPROCESSING

A total of 8 healthy volunteers (4 males and 4 females, aged 26.5 ± 4.8 years, ranging from 20 to 35 years old) were recruited in this study. To ensure diversity for cross-subject evaluation, subjects with an equal number of left-handed and right-handed individuals and diverse physical characteristics were enrolled. Their detailed demographic information is summarized in Table I.

The experimental system was built using a customized hybrid sensing band, which consists of 4 sensing nodes. Each node integrates a dry surface electromyography (sEMG) electrode and a single-element AUS transducer. The sensing band was uniformly worn on the forearm of each subject to monitor the major muscle groups related to wrist and finger coordinated movements. The schematic diagram of the device placement is shown in Fig. 3(a). In this study, six

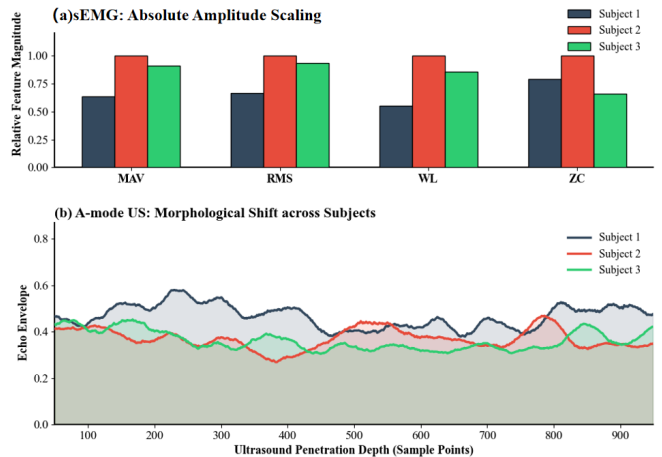


Fig. 1. Schematic illustration of inter-subject signal variability when performing the identical “scissors” gesture. (a) Amplitude scaling drift in surface electromyography (sEMG) signals across subjects. (b) Morphological and depth variations of deep muscle tissues captured by AUS echoes.

TABLE I
DEMOGRAPHIC INFORMATION OF ALL SUBJECTS

Subj.	Sex	Age (yr)	Height (cm)	Weight (kg)	Hand
S1	M	21	178	70	R
S2	F	32	162	50	R
S3	M	35	182	88	R
S4	F	24	165	58	R
S5	M	20	175	75	L
S6	F	28	160	48	L
S7	M	26	170	80	L
S8	F	26	168	62	L

representative functional gestures were defined, as illustrated in Fig. 3(b): thumb flexion (TF), index flexion (IF), pinch (PH), fist (FT), peace sign (PS) and open hand (HO).

The data acquisition protocol for each gesture was standardized as follows: a 5-second preparation phase, a 3-second steady holding phase, and a 5-second rest phase. Each subject performed 8 repeated trials for each gesture. The sampling frequencies of the sEMG signals and AUS signals were set to 1000 Hz and 20 Hz, respectively. The complete data set and its data structure have been open source and are available at <https://zenodo.org/me/requests/8abc8599-d259-43ac-8af4-6881b4d70189>.

To extract generalizable representations from high-dimensional, non-stationary raw physiological data, this study adopts a dual-path feature extraction pipeline, optimized for electrophysiological and muscle morphological offsets respectively.

- 1) **Standard sEMG Time-Domain Descriptors:** This study employs four classic and widely recognized time-domain statistical features in the field of EMG control to capture the activation intensity of neuromuscles. For each 200 ms analysis window, the model calculates the Mean Absolute Value (MAV), Root Mean Square (RMS), Waveform Length (WL), and Zero Crossing (ZC) of the 4-channel signals [18]. MAV and RMS provide estimates of the amplitude and energy of

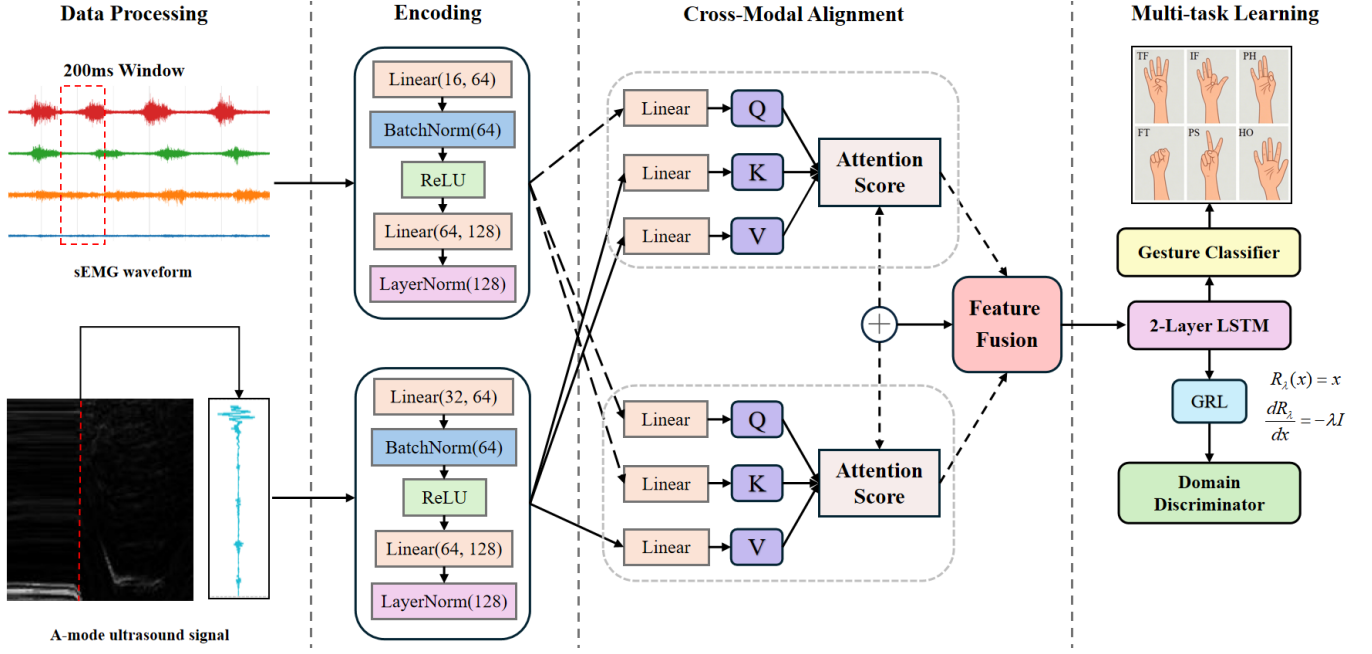


Fig. 2. Detailed architecture of the SIAF-Net framework. The pipeline sequentially executes feature encoding with baseline calibration, bi-directional cross-modal alignment for modality interaction, and joint adversarial learning via a Gradient Reversal Layer (GRL) to achieve subject-invariant gesture recognition.

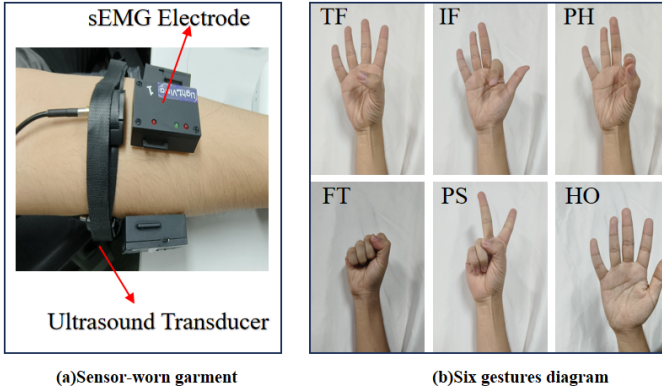


Fig. 3. Experimental setup and definitions of the functional gestures. (a) The customized hybrid sensing band, equipped with sEMG electrodes and an AUS transducer, worn on the forearm. (b) The six representative functional gestures targeted for robotic control: thumb flexion (TF), index flexion (IF), pinch (PH), fist (FT), peace sign (PS), and hand open (HO).

muscle contraction; WL reflects the cumulative complexity and force generation rate of the signal; ZC characterizes the firing frequency of motor units. This 16-dimensional (4 channels $\times$ 4 features) low-dimensional feature combination exhibits higher robustness than raw waveforms when dealing with inter-individual impedance drift.

2) AUS Spatial Binning Alignment: To address the feature misalignment problem of AUS caused by individual differences, this study introduces the spatial binning strategy proposed in [19], instead of traditional point-to-point sampling. We divide the 128-point A-mode echo envelope processed by Hilbert transform into 8 equal-length spatial subdomains (bins) along the depth direction. By calculating the average

acoustic energy in each bin (4 channels $\times$ 8 bins), we convert local reflection peaks that are highly susceptible to fat layer thickness interference into macroscopic descriptors of muscle tissue deformation. This dimensionality reduction mechanism preserves the integrity of deep muscle morphological changes and achieves preliminary alignment of individual anatomical differences at the physical level.

3) Subject-Level Normalization: Before entering the alignment network, all 48-dimensional joint features are standardized through intra-subject global Z-score normalization. This step aims to eliminate the “magnitude noise” caused by differences in muscle volume and baseline electrical level among different subjects, ensuring that the multi-modal fusion process focuses on dynamic contraction patterns rather than individual static anatomy.

III. METHODS

The proposed unified framework, termed the Subject-Invariant Alignment and Fusion Network (SIAF-Net), is designed to extract robust neuromuscular representations from heterogeneous sEMG and AUS modalities. As illustrated in Fig.2, the architecture comprises three functional hierarchies: feature encoding with baseline calibration, bi-directional cross-modal alignment, and joint adversarial-temporal modeling.

A. Feature Encoding and Baseline Calibration

To mitigate the shifts and signal amplitude scaling identified in Fig. 1, we operate on engineered statistical feature spaces rather than raw high-frequency waveforms. For each temporal window t , the input descriptors are defined as

$\mathbf{x}_e^{(t)} \in \mathbb{R}^{16}$ (sEMG statistical descriptors) and $\mathbf{x}_u^{(t)} \in \mathbb{R}^{32}$ (AUS spatial bin energies).

Traditional normalization techniques, such as BatchNorm, often fail in cross-subject scenarios because their dependency on global batch statistics is susceptible to corruption by significant individual variances. Consequently, we introduce Layer Normalization (LN) within each MLP encoder branch to perform sample-wise physiological baseline calibration:

$$\mathbf{f}_m^{(t)} = \text{LN}\left(\text{MLP}_m(\mathbf{x}_m^{(t)})\right), \quad m \in \{e, u\} \quad (1)$$

where $\mathbf{f}_e^{(t)}, \mathbf{f}_u^{(t)} \in \mathbb{R}^d$ ($d = 128$) represent the calibrated latent embeddings. The LN ensures that feature magnitudes remain standardized across varying skin impedances and fat thicknesses, providing a consistent manifold for subsequent semantic alignment.

B. Bi-directional Cross-Modal Alignment

Existing fusion paradigms typically utilize static concatenation $\mathbf{f}_{fused} = [\mathbf{f}_e; \mathbf{f}_u]$, which implicitly assumes both modalities are equally reliable across all individuals. To account for unconstrained anatomical variations, we propose a Bi-directional Cross-Attention module to dynamically retrieve consistent semantic correlations. Specifically, one modality acts as the Query (Q) to interrogate the other's Key (K) and Value (V):

$$\mathbf{Q}_e = \mathbf{W}_{q,e}\mathbf{f}_e, \quad \mathbf{K}_u = \mathbf{W}_{k,u}\mathbf{f}_u, \quad \mathbf{V}_u = \mathbf{W}_{v,u}\mathbf{f}_u \quad (2)$$

To suppress noise originating from anatomical misalignments in the AUS modality, the alignment weights are computed via a gated attention mechanism:

$$\mathbf{A}_{e \rightarrow u} = \sigma\left(\frac{\mathbf{Q}_e \mathbf{K}_u^\top}{\sqrt{d}}\right) \quad (3)$$

where σ denotes the sigmoid activation. The sEMG-aligned feature is then formulated as $\hat{\mathbf{f}}_e = \mathbf{f}_e + \mathbf{A}_{e \rightarrow u} \mathbf{V}_u$. Through a symmetric operation, the AUS-aligned feature $\hat{\mathbf{f}}_u$ is obtained. The multimodal representation at time t is defined as the average of these aligned embeddings:

$$\mathbf{f}_{fused}^{(t)} = \frac{1}{2} \left(\hat{\mathbf{f}}_e^{(t)} + \hat{\mathbf{f}}_u^{(t)} \right) \quad (4)$$

This mechanism allows the model to adaptively prioritize the stable electrophysiological triggers of sEMG when the morphological signatures of AUS suffer from depth-related shifts in held-out target users.

C. Joint Adversarial Learning and Optimization

To ensure that the extracted features are truly subject-invariant, we integrate a Domain Adversarial Neural Network at the output of a 2-layer LSTM temporal encoder. The temporal features $\mathbf{h} = \text{LSTM}(\{\mathbf{f}_{fused}^{(1)}, \dots, \mathbf{f}_{fused}^{(T)}\})$ are fed into a dual-head structure.

1) *Adversarial Alignment*: We introduce a Gradient Reversal Layer, which acts as an identity mapping during forward propagation but reverses the gradient during back-propagation:

$$R_\lambda(\mathbf{h}) = \mathbf{h}; \quad \frac{dR_\lambda}{d\mathbf{h}} = -\lambda \mathbf{I} \quad (5)$$

A domain discriminator $D(\cdot)$ minimizes the domain loss L_{dom} to distinguish between Source and Target subjects, while the feature extractors are trained to maximize this loss, effectively stripping away subject-specific ‘‘anatomical fingerprints.’’

2) *Unified Objective Function*: The entire framework is optimized through a multi-objective loss function:

$$L_{total} = L_{gest} + \gamma L_{dom} + \beta L_{sim} + \eta L_{ent} \quad (6)$$

The physical implications of the loss components are as follows:

- L_{gest} : Negative log-likelihood loss for gesture classification to ensure intentional decoding accuracy.
- $L_{sim} = \|\mathbf{f}_e - \mathbf{f}_u\|_2^2$: Modality Consistency Loss, enforcing the heterogeneous sensors to converge on a shared semantic manifold of the same muscle activity.
- L_{ent} : Entropy Minimization Loss applied to the target domain, designed to encourage confident decision boundaries and predictions on held-out target subjects.

IV. EXPERIMENTS

To comprehensively quantify the performance of SIAF-Net and investigate the synergistic relationship between multimodal fusion and cross-subject alignment, we designed a rigorous evaluation framework comprising the following experimental scenarios.

A. Verification Protocols and Scenarios

To simulate real-world robotic deployment, we utilize a Leave-One-Subject-Out (LOSO) protocol across all cross-subject experiments. In each fold, one subject's data is reserved as the held-out target domain, while the remaining subjects constitute the source domain. Two primary scenarios are defined:

- 1) **Cross-Subject Single-modal Baseline**: This scenario serves as the performance lower-bound. By activating only the sEMG or AUS branch individually, we reveal the independent generalization capability of electrophysiological and morphological features for held-out target subjects.
- 2) **Cross-Subject Multi-modal Alignment**: As the core task of this study, we evaluate SIAF-Net and representative fusion paradigms to verify the mutual enhancement effect between inter-modal alignment and subject-invariant representation learning.

In each LOSO fold, one subject served as the target domain, and the remaining seven subjects formed the labeled source domain. Target-subject gesture labels were removed and never used for training, validation, model selection, or hyperparameter tuning. Target-domain samples were used

TABLE II
COMPREHENSIVE PERFORMANCE COMPARISON UNDER LOSO
SETTING

Category	Mod.	Acc. (%)	MF1	SD (%)
<i>Unimodal Methods</i>				
sEMG-MLP	E	45.62	0.42	12.35
AUS-MLP	U	32.14	0.28	16.48
<i>Basic Fusion Methods</i>				
Static Concatenation[16]	E+U	51.45	0.48	11.20
Transformer Fusion[17]	E+U	62.58	0.59	9.87
<i>Domain Adaptation Methods</i>				
TCA[20]	E+U	55.40	0.51	10.56
MMD[21]	E+U	58.72	0.54	10.12
DANN[12]	E+U	68.35	0.64	9.15
MMCANet+DANN	E+U	71.24	0.68	8.54
Proposed Method				
SIAF-Net	E+U	81.42	0.78	7.65

only as unlabeled inputs for adversarial alignment and entropy minimization. Thus, the offline evaluation represents unsupervised cross-subject adaptation without labeled target-user gesture calibration. After training under this label-free adaptation protocol, the online validation required only a 2-s resting baseline normalization before inference, and no labeled target-user gesture trials were collected.

B. Baselines and Fairness Justification

Given that cross-subject research combining AUS and sEMG is relatively nascent, we implemented representative algorithmic paradigms from the literature and adapted them to the LOSO environment to provide a rigorous benchmark:

- **sEMG-MLP / AUS-MLP:** Single-modal baselines utilizing 16-D and 32-D statistical/spatial features, respectively, to establish a performance lower-bound for individual sensors.
- **Static Concatenation:** A replication of the core paradigm in early multimodal studies, utilizing simple feature-level concatenation without any dynamic alignment or adversarial regularizer.
- **Transformer Fusion:** It incorporates multilevel self-attention for modality interaction but excludes the domain adversarial branch.
- **MMCANet + DANN:** A strong recent multimodal baseline that integrates the attention backbone of [17] with a DANN-based alignment branch. This model serves as a strong reference baseline that serializes existing independent fusion and alignment technologies.
- **SIAF-Net (Ours):** The proposed unified framework, which concurrently activates bi-directional cross-modal attention and domain adversarial alignment within a single optimization loop to exploit their mutual enhancement.

To ensure a fair comparison, all baselines used the same LOSO splits, preprocessing, feature extraction, temporal window length, evaluation metrics, optimizer, learning-rate schedule, epochs, and source-target protocol. Multimodal

methods used the same 16-D sEMG descriptors and 32-D AUS spatial-bin features. Transformer Fusion retained the fusion backbone but removed adversarial and modality-consistency losses. MMCANet+DANN used the same GRL-based discriminator and unlabeled target-domain setting as SIAF-Net. Target-subject gesture labels were never used for training or hyperparameter selection.

C. Performance Metrics

To comprehensively evaluate the proposed framework, we define two sets of metrics targeting offline algorithmic robustness and online robotic control stability, respectively.

1) Offline Classification Metrics: For the 6-class gesture classification task, we report accuracy, Macro-F1, and the standard deviation across LOSO folds. Macro-F1 is used to reduce the potential bias caused by class imbalance by averaging class-wise F1 scores.

2) Online HIL Control Metrics: To quantitatively assess the online closed-loop interaction quality during the continuous robotic deployment (detailed in Sec. VI), we define five system-level metrics:

- **Online Accuracy (%):** The raw frame-by-frame classification accuracy across the continuous data stream.
- **Command Jitter Rate (%):** The frequency of rapid, unintended prediction switches between adjacent inference frames during a steady-state phase.
- **Task Success Rate (%):** It is defined as the ratio of successful trials in which the subject maintains the target posture with high-confidence predictions for the required 3-second window.
- **Mean Confidence Score:** The average softmax probability of the predicted class, serving as the threshold criterion for the safety filter.
- **Avg. Response Delay (s):** The end-to-end latency measured from the user’s initial muscle contraction to the completion of the manipulator’s mechanical response.

D. Hyperparameter Settings and Implementation

The framework is implemented using PyTorch 2.1 and executed on an NVIDIA GeForce RTX 5080 GPU.

- **Optimization Strategy:** We employ the AdamW optimizer with a weight decay of 10^{-2} to mitigate overfitting. The initial learning rate is set to 10^{-3} , which is smoothly decayed via a Cosine Annealing scheduler over 100 epochs to ensure stability during adversarial training.
- **Adversarial Control:** The gradient reversal factor α is evolved based on the training progress p :

$$\alpha = \frac{2}{1 + \exp(-10 \cdot p)} - 1 \quad (7)$$

- **Loss Weights:** Through grid search, the coefficients for the composite objective function are determined as $\gamma = 0.5$ (domain alignment), $\beta = 0.1$ (modality consistency), and $\eta = 0.05$ (entropy minimization).

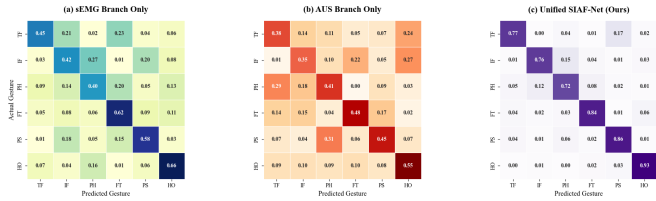


Fig. 4. Confusion matrices demonstrating the progressive improvement in classification performance under the cross-subject setting. (a) sEMG-only baseline, (b) AUS-only baseline, and (c) the proposed multimodal SIAF-Net framework.

V. RESULTS AND ANALYSIS

A. Classification performance

As shown in Table II, we compare SIAF-Net with various unimodal models and mainstream multi-modal fusion paradigms. The experimental results support the following observations:

1) **Limitations of Unimodal Models:** In the stringent leave-one-subject-out (LOSO) task, unimodal models exhibit high instability. Specifically, the accuracy of AUS-MLP is only 32.14% with a standard deviation as high as 16.48%. For held-out target users, the spatial misalignment of ultrasound echoes causes traditional classifiers to show limited generalization.

2) **Superiority of the Fusion Framework:** The proposed SIAF-Net (Ours) achieves an average accuracy of 81.42% under label-free cross-subject adaptation without labeled target-user gesture calibration. SIAF-Net achieves the highest average accuracy among all compared methods and outperforms the strong reference baseline MMCANet + DANN by 10.18%. This advantage stems from the fact that our unified framework not only completes feature stacking but also performs adaptive cross-modal feature alignment at the physiological semantic level through the bi-directional cross-attention mechanism.

3) **Improved Stability:** SIAF-Net achieves the lowest standard deviation (7.65%) across all experiments. This suggests that adversarial alignment helps reduce subject-specific variations related to body composition and force-generation habits, enabling the model to learn more transferable neuromuscular representations.

TABLE III
COMPACT ABLATION RESULTS UNDER LOSO. CA DENOTES CROSS-ATTENTION AND DA DENOTES DOMAIN-ADVERSARIAL ALIGNMENT.

Method	CA	DA	Acc. (%)
<i>Modality masking</i>			
sEMG retained, AUS masked	—	—	52.14 $\pm$ 9.42
AUS retained, sEMG masked	—	—	43.85 $\pm$ 14.12
<i>Component ablation</i>			
Base concatenation	×	×	45.12 $\pm$ 11.54
Cross-attention only	✓	×	52.48 $\pm$ 9.87
Adversarial only	×	✓	58.75 $\pm$ 8.92
SIAF-Net	✓	✓	81.42 $\pm$ 7.65

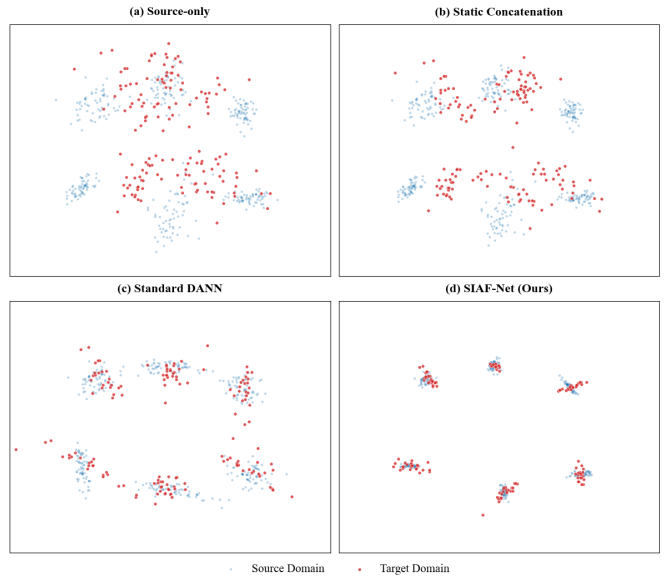


Fig. 5. The t-SNE visualizations of the learned high-level feature spaces across different cross-subject alignment strategies. (a) Source-domain-only baseline without alignment, (b) Static Concatenation, (c) Standard DANN without dynamic attention, and (d) the proposed unified SIAF-Net. Source domain samples are plotted in blue, and unlabeled target-domain samples in red.

B. Ablation Study and Performance Evaluation

We further performed modality-masking and component-ablation analyses (Table III). For modality masking, the complete SIAF-Net was trained once with both inputs; during evaluation, the suppressed modality embedding was replaced by a zero vector, and all parameters remained unchanged. Retaining only sEMG gives 52.14% $\pm$ 9.42%, whereas retaining only AUS gives 43.85% $\pm$ 14.12%. This comparison suggests that sEMG provides more transferable activation cues under the tested LOSO setting. In contrast, AUS alone is more sensitive to depth-related anatomical shifts, probe-contact variation, and differences in soft-tissue morphology across users.

The component ablation shows that simple feature concatenation is insufficient for cross-subject transfer, reaching only 45.12% $\pm$ 11.54%. This result indicates that placing the two feature vectors in a shared input space does not by itself resolve inter-modal heterogeneity or subject-dependent distribution shifts. Cross-attention alone improves the result to 52.48% $\pm$ 9.87%, suggesting that explicit inter-modal interaction helps reduce local modality mismatch. However, without a subject-alignment constraint, the fused features can still retain user-specific patterns.

DA alone reaches 58.75% $\pm$ 8.92%, confirming the value of adversarial subject alignment for reducing target-user distribution mismatch. The complete SIAF-Net achieves 81.42% $\pm$ 7.65%, which is higher than either component alone. This trend supports the design motivation of coupling CA and DA: cross-modal alignment provides a more consistent multimodal feature space for adversarial learning, while DA regularizes the fused representation toward subject-invariant gesture semantics.

C. Synergistic Analysis

To further intuitively demonstrate the manifestation of this “mutual enhancement” in practical action recognition, we compare the confusion matrices of each branch in Fig. 4.

As shown in Fig. 4(a), the pure sEMG branch exhibits severe misclassification in recognizing fine fingertip movements. Fig. 4(b) shows that the pure AUS branch has almost fragmented decision boundaries due to echo peak shifts. In SIAF-Net (Fig. 4(c)), this confusion is largely corrected. Using electrophysiological triggers as “semantic guidance”, the model reduces the physical drift of AUS features. The reduction in off-diagonal misclassification noise indicates that the framework can provide more consistent perception capabilities for users with different physical characteristics.

D. Feature Space Visualization

To verify the domain-invariant, highly discriminative high-level semantic representations generated by our method, we visualize the features learned by different cross-subject recognition methods in Fig. 5. The representations obtained by all methods are projected onto a two-dimensional space via the t-distributed Stochastic Neighbor Embedding (t-SNE) algorithm.

Compared with other cross-subject alignment methods, as shown in Fig. 5(a), the representation space of the pure source-domain method exhibits low inter-class discriminability, and there is obvious physical separation between the data distributions of the source domain (blue) and the target domain (red). Although the static concatenation method reduces the domain distance to a certain extent, the feature distribution still shows clear “subject dependency”, making it difficult to form effective classification boundaries.

As can be seen from Fig. 5(c), the classic DANN module can initially extract domain-invariant features from multi-modal data, achieving a high degree of overlap between red and blue sample points. However, due to the failure to address inter-modal heterogeneity conflicts, the classification boundaries remain ambiguous.

Fig. 5(d) shows the final projected feature space of the proposed SIAF-Net. Compared with the standard adversarial module in Fig. 5(c), samples of the same gesture are more tightly clustered into 6 independent manifold clusters, while different gesture states show clearer separation, enabling more robust cross-subject intent decoding.

VI. ROBOTIC CONTROL SYSTEM EVALUATION

To verify the deployment capability of SIAF-Net in real-world robotic tasks, we constructed an end-to-end hardware-in-the-loop (HIL) validation platform. This system integrates a wearable hybrid sensing band, a dedicated inference node based on an RTX 5080 GPU, and a 5-degree-of-freedom (DoF) dexterous manipulator.

A. Sequential Analysis of Continuous Intent Mapping

Fig. 6 illustrates the continuous control process of the system for a held-out target subject whose gesture labels were not used during training.



Fig. 6. Continuous control process of the robotic manipulator for a held-out target subject. (I) Rest phase: fully open posture with high resting-state confidence; (II) Transition phase: temporary confidence fluctuation due to anatomical dynamic shift; (III) Stable state: confidence rebounds to 98.2% with stable manipulator posture.

TABLE IV
QUANTITATIVE ONLINE PERFORMANCE SUMMARY ON HELD-OUT
TARGET SUBJECTS

Metric	sEMG Only	AUS Only	SIAF-Net (Ours)
Online Accuracy (%)	42.41 $\pm$ 8.65	38.65 $\pm$ 12.30	82.14 $\pm$ 5.12
Command Jitter Rate (%)	12.85 $\pm$ 4.20	24.52 $\pm$ 6.85	1.52 $\pm$ 0.45
Task Success Rate (%)	62.50 $\pm$ 15.2	37.50 $\pm$ 18.4	87.50 $\pm$ 7.50
Mean Confidence Score	0.72 $\pm$ 0.08	0.61 $\pm$ 0.12	0.96 $\pm$ 0.02
Avg. Response Delay (s)	0.85 $\pm$ 0.12	0.92 $\pm$ 0.18	1.04 $\pm$ 0.15

1) Phase I (Rest): In the user’s relaxed state, the model outputs high resting-state confidence, and the manipulator maintains a stable fully open posture.

2) Phase II (Transition): At the moment of action triggering, the subject’s muscle morphology undergoes pronounced deformation. Due to the dynamic shift of anatomical structures, the recognition confidence exhibits a temporary fluctuation (78.5%). This process illustrates that the cross-modal attention module performs online feature alignment during gesture evolution.

3) Phase III (Stable State): As the action enters a steady state, the bi-directional attention mechanism successfully locks onto cross-modal features, and the recognition confidence rebounds and stabilizes at 98.2%. Experiments demonstrate that even under signal nonlinear variations caused by cross-subject differences, the proposed framework can still provide smooth and intuitive control feedback.

B. Quantitative Online Performance and Stability Analysis

We statistically analyzed the online performance of the system across 8 LOSO experimental folds. Each subject performed a cumulative 15-minute continuous online test. The test data are summarized in Table IV. Based on the system-level metrics defined in Sec. IV-C, several key observations are made:

- **Command Stability:** For held-out target users, single-modal systems suffer from frequent misclassification “fluctuations” in the command stream due to their inability to handle distribution shifts, with command jitter rates of 12.85% and 24.52%, respectively. In contrast, SIAF-Net achieves a low jitter rate of $1.52\% \pm 0.45\%$. This level of stability is crucial for extending the mechanical lifespan of robotic actuators and ensuring safe physical human-robot interaction.
- **Decision Confidence:** SIAF-Net achieves an average confidence score of 0.96 ± 0.02 . This high confidence

ensures that the manipulator executes actions only when electrophysiological triggers and physical deformations reach semantic consensus, greatly reducing the risk of false triggering (the task success rate reaches 87.50%).

C. Online Deployment Details and Latency Analysis

To address the non-causal global view issue in online sampling, we adopted a causal initialization strategy based on resting baseline normalization. Within the first 2 seconds after the user wears the sensing band, the system records resting-period signals to estimate the mean and variance specific to that user. These parameters are fixed for subsequent online inference, and no labeled target-user gesture trials are collected. The measured end-to-end response delay of the system is $1.04 \text{ s} \pm 0.15 \text{ s}$. SIAF-Net's core algorithmic processing, including feature extraction and inference, consumes 12.4 ms. The remaining delay is hardware and protocol bound, comprising the obligatory 200 ms sEMG analysis window, 50 ms AUS sampling interval, and 770 ms of mechanical actuation delay inherent to our specific tendon-driven manipulator and serial communication. Therefore, the current validation demonstrates online closed-loop feasibility rather than high-speed manipulation.

VII. CONCLUSIONS AND FUTURE WORKS

This paper proposes SIAF-Net, an integrated framework for label-free cross-subject sEMG-AUS gesture recognition. Instead of requiring labeled target-user gesture calibration, the proposed framework combines bidirectional inter-modal alignment with adversarial subject alignment to learn more transferable neuromuscular representations. In the online robotic validation, the only user-specific operation is a short 2-s resting baseline normalization before inference. By leveraging the electrophysiological stability of sEMG as a "semantic anchor" to dynamically calibrate AUS morphological shifts via bi-directional cross-attention, SIAF-Net achieves a robust $81.42\% \pm 7.65\%$ accuracy under LOSO validation. Our method reduces inter-subject performance fluctuations and supports online closed-loop robotic validation under the tested protocol. Future work will expand the subject cohort, improve probe robustness with flexible ultrasound patches, extend the framework to continuous hand kinematics estimation, and reduce the current 1.04 s end-to-end response delay through embedded deployment and faster actuation hardware.

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